

SC

No. of printed pages: 02

[A5]

SARDAR PATEL UNIVERSITY
BCA SEM-VI EXAMINATION 2015
INFORMATION SECURITY (US06EBCA01)

Time: 10:30 a.m. to 12:30 p.m.

Date: 10/04/2015

Total Marks: 70

Q.1 Multiple Choice Questions

[10]

- 1 Which action is that compromises the security of information owned by an organization?
a) Security mechanism b) Security Attack c) Service attack d) Threat
- 2 Which category of Active attack involves the passive capture of a data unit and its subsequent retransmission?
a) Replay b) Masquerade c) Traffic analysis c) None of these
- 3 Traffic analysis is the type of _____ attack.
a) Active Attack b) Passive attack c) Service attack d) both A & B
- 4 Cryptography is also known as _____.
a) plain text writing b) stream writing c) secret writing d) None of above
- 5 The _____ performs various substitutions and transformations on the plaintext.
a) plaintext b) Encryption algorithm c) secret key d) cipher text
- 6 The process of attempting to discover the plaintext or key is known as _____.
a) Cryptanalysis b) Cipher text c) Stream cipher d) Block cipher
- 7 _____ is a person who is authorized for access but misuses his or her privileges.
a) Misfeasor b) Masquerader c) Clandestine user d) None of these
- 8 Infection mechanism is also referred as _____.
a) Infection spread b) Infection vector c) Infection propagation d) All of above
- 9 _____ is a server that creates a secure tunnel connection.
a) DNS b) VPN c) Tunnel d) authentication
- 10 Stateful firewall maintains a _____, which is a list of active connection.
a) Routing table b) Bridging table c) State table d) Connection table

Q.2 Write Short Questions. [Any 10] [20]

- 1 Differentiate between **Active & Passive Attacks**.
- 2 Define **NonRepudiation**.
- 3 Define **Security Attack** and **Security Mechanism**.
- 4 Which are two requirements for **secure use of symmetric encryption**?
- 5 List the three dimensions for classification **Cryptographic System** and define any one.
- 6 Define **Encryption** and **Decryption Algorithm**.
- 7 Define : **Polymorphic virus & Metamorphic virus**
- 8 Discuss **third generation of Antivirus approach**
- 9 Define: **Trigger** and **Payload**.
- 10 Define **Digital signature**.
- 11 Discuss **DMZ**.
- 12 List **types of Firewalls** and define any one.

Q.3(A) Write a detail note on Authentication, Access Control and Data Confidentiality. [06]
(B) Define Active Attack. List all and explain any one in detail [04]

OR

Q.3(A) Explain the Network Security Model in detail? [06]
(B) Discuss Release of Message and Traffic Analysis. [04]

Q.4(A) Detail out Public Key Encryption Structure. [06]
(B) Write a note on types of attack on Encrypted Message. [04]

OR

Q.4(A) List and define six ingredients of Public-key Encryption Scheme. [06]
(B) Write a detail note on Applications of Public key Cryptosystem [04]

Q.5(A) Discuss Virus Classification by target and by concealment strategy. [06]
(B) Write a note on Structure of Virus. [04]

OR

Q.5(A) Discuss Anti-virus approach in detail. [06]
(B) Define the following: a) Adware, b) Trojan horse, c) Logic Bomb, d) Key Loggers [04]

Q.6 Write a detail note on Need, Characteristics and Limitations of Firewall. [10]
OR

Q.6 List the services of Digital Signature and explain in detail. [10]
